Gabriel Cirker

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EDUCATION

University of Michigan, *Ann Arbor*.....Expected May 2028

- **Double major:** Physics and Mathematics
- **Current courses:** MATH 463: Mathematical Biology; PHYSICS 411: Computational Physics

St. John's University, prior study in Physics and Mathematics; GPA 4.00 through Fall 2025

RESEARCH AND PROJECTS

Immune Activation and Self-Tolerance

MATH 463 three-person group project.....Fall 2026

- Coauthored a report on conventional T-cell, regulatory T-cell, and IL-2 dynamics using an adapted three-state model from the literature.
- Group analysis used stiff ODE integration, parameter sweeps, and SINDy with separate training, validation, and test trajectories; recovered all 12 model terms in a clean-data baseline.

Project and report

HIV mRNA Structure and Energy Modeling

St. John's University.....Jan - Jun 2026

- Analyzed HIV RNA in R to examine associations between sequence length, structural features, motif placement, and predicted minimum free energy.
- Developed structure-derived predictors, regression comparisons, and diagnostic plots for a reproducible analysis workflow.

Computational Biomechanics of Rowing

Independent project | Project and report

- Developed a physics-based model of mechanical energy transfer from rowing strokes to boat motion, accounting for losses and efficiency.
- Used Python and NumPy/SciPy for simulation and parameter sweeps; documented modeling assumptions, results, and limitations in a technical report.

Lightweight Bridge Modeling

Independent project | Project and report

- Compared equal-mass beam geometries using Euler-Bernoulli theory, numerical deflection calculations, and buckling analysis.

GPU Compute Rig

Independent hardware project

- Designed, built, and tuned an 8-GPU system using power and thermal monitoring to evaluate performance and efficiency.

SELECTED EXPERIENCE AND RECOGNITION

Assistant Rowing Coach

Friends of Port Rowing.....Jun 2023 - Aug 2025

Coached technique, biomechanics, and race strategy; supported training and athlete development.

First place, Freshman Engineering Bridge Competition

University of San Diego.....May 2024

TECHNICAL SKILLS

Python, R | NumPy, SciPy, pandas, Matplotlib, tidyverse, ggplot2 | Git, Jupyter, LaTeX

Numerical simulation, regression, parameter sweeps, scientific visualization, reproducible analysis